

Operator's Manual

3T 8 Channel High Res Brain Array for GE 3T MR System



Manufacturer



Invivo Corporation
3545 SW 47th Avenue
Gainesville, FL 32608-7691 U.S.A.
352-336-0010

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GE Healthcare



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Proper performance of this coil is guaranteed only while the coil is being used on the GE 3T MR system (hardware/software level) specified at the time of purchase. Upgrades or other modifications to the system software and/or hardware may affect compatibility. Prior to upgrading your system, please contact your GE Healthcare representative to discuss coil compatibility issues. Failure to do so may void your warranty.

Medical Device Directive

These products conform with the requirements of council directive 93/42/EEC concerning medical devices, when they bear the following CE Mark of Conformity:

European Representative

GBM Authorised Representative Ltd.
The White House, 2 Meadow
Godalming, Surrey GU7 3HN
United Kingdom



 Attention, Consult Accompanying Documents	NOTICE: This equipment shall be transported and stored under the following conditions: <u>Shipping</u> 1. Ambient temperature range of -40°C to +65°C. 2. Relative humidity range of 10% to 100%, non-condensing. <u>Storage</u> 1. Ambient temperature range of +10°C to +40°C. 2. Relative humidity range of 10% to 90%, non-condensing. Atmospheric pressure range of 500 hPa to 1060 hPa.
 Type BF Equipment	
 Class II Equipment	

WARNING: This product contains chemicals, including lead, known to the state of California to cause birth defects or other reproductive harm. **Wash hands after handling.**

Caution:
Federal law restricts this device to sale, distribution and use by or on the order of a physician.

INTRODUCTION

This manual describes the safety precautions, features, use and care of the Invivo

- HD 8 Ch HiRes Brain Array (Receive Only)
- PETMR 8Ch HiRes Brain Array (Receive Only)

In this manual both coils are referred as 3T 8Ch Brain Coil.

Please review this manual thoroughly before using the device.

If you have any questions or comments regarding this manual, or need any assistance with the use of the product, contact information is located at the end of this document.

COMPATIBILITY

The Invivo 3T 8Ch Brain Coil is compatible with GE 3T MR Systems.

***NOTE:** PETMR 8Ch HiRes Brain Array differs from HD 8Ch Brain Array in following

1. The coils have different **coil id**.
2. The **mirror assembly** design is different in order to reduce the height of the mirror to fit in PET/MR bore. Use mirror assemblies only provided with PETMR 8Ch HiRes Brain Array.

* Refer SIGNA PET/MR Service Manual Direction (For MCQA): 5670014
and SIGNA PET/MR Operator Manual (for PET/MR specific 8Ch Brain coil patient setup)

EXPLANATION OF SYMBOLS



Medical Device Directive - These products conform with the requirements of council directive 93/42/EEC concerning medical devices, when they bear the CE Mark of Conformity.



Attention, Consult Accompanying Documents.



Type BF Equipment



Class II Equipment



Do not cross or loop cables. Arcing and patient burns could result.



Waste in Electrical and Electronic Equipment



A hazard exists which could cause serious injury.

Manufacturer



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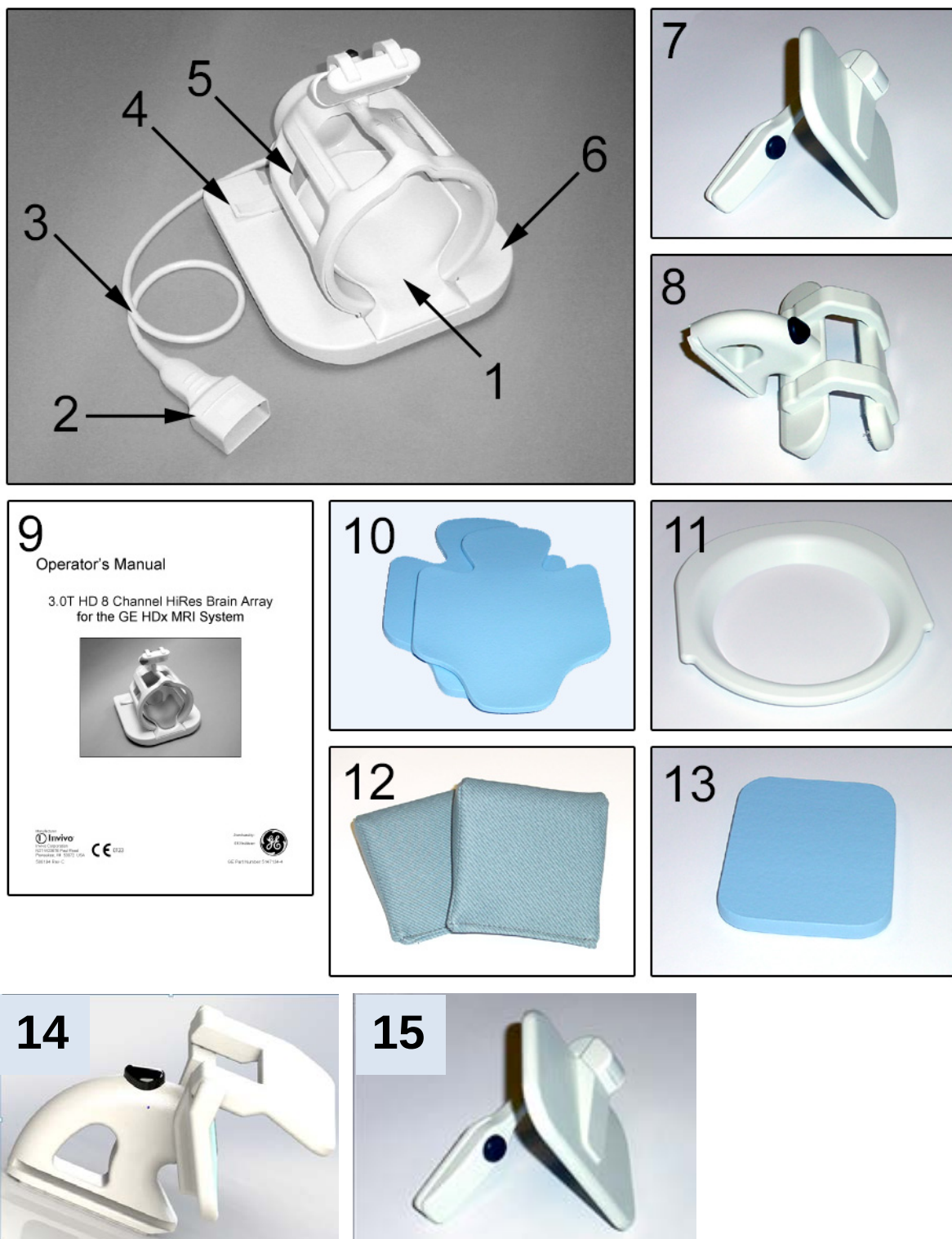


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CHAPTER 1 - 3T 8Ch Brain Coil CONTENTS

The 3T 8Ch Brain Coil consists of the following parts. Please inspect upon receipt to make sure all parts have arrived and are in good order. Use this guide to refer to part names throughout this manual.



- | | | |
|-------------------|----------------------------|------------------------|
| 1. Head Support | 6. Base | 11. Phantom Positioner |
| 2. Coil Connector | 7. Superior Mirror | 12. Wedge Pads |
| 3. Coil Cable | 8. Inferior Mirror | 13. Spacer Pad |
| 4. Locking Lever | 9. Operator's Manual | |
| 5. Window | 10. Thick & Thin Head Pads | |

For PETMR 8Ch HiRes Brain Array (Receive Only), instead of item #7 & #8 PETMR Mirrors #14 & #15 should be part of content.

- 14. Inferior Mirror for PETMR
- 15. Superior Mirror for PETMR

CHAPTER 2 - SAFETY

2-1 Training

This manual contains detailed information regarding the set-up, positioning and use of the Invivo Corporation coil. The instructions must be read carefully and thoroughly before attempting to scan patients with this coil.

2-2 Quality Assurance

The procedure described in the Quality Assurance chapter of this manual should be performed upon receipt of the coil to establish a baseline of coil performance. The procedure should be repeated at regular intervals.

2-3 Indications

The coil is indicated for use, on the order of a physician, in conjunction with a MR scanner as an accessory to produce images of the head and brain.

2-4 Contraindications

The operator should be aware of the following contraindications for use related to the strong magnetic field of the MR system:

 Scanning is contraindicated for patients who have electrically, magnetically or mechanically activated implants (cardiac pacemakers, for example). The magnetic and electromagnetic fields produced by the MR system and coil may interfere with the operations of these devices.

 Scanning patients with intracranial aneurysm clips is contraindicated.

2-5 Precautions

Precautions should be taken when scanning patients with the following conditions:

 Greater than normal potential for cardiac arrest

 An increased likelihood for developing seizures or claustrophobia

 Unconscious, heavily sedated, or confused patients

 Inability to maintain reliable communications

2-6 Cautions

The following general warning statements apply to scanning with a magnetic resonance system. For further details, review the warnings in your system Operator's Manual.

-   Do not cross or loop cables. Arcing and patient burns could result. Route cables out of the magnet so that they do not touch the patient.
-  Assure that the patient is not touching the bore. If necessary, place pads between the patient and the surface of the bore.
-  If the patient complains of warming, tingling, stinging, or similar sensations, promptly stop the scan procedure, examine the patient, and contact the responsible physician before continuing the procedure. Pay special attention to very young, sedated, or other compromised patients who may not be able to communicate effectively.
-  Patients with implanted ferromagnetic metal should not be scanned because the magnetic field may interact with implanted surgical clips or other ferromagnetic materials.
-  Persons with cardiac pacemakers or other implanted electronic devices should not enter the magnetic field zone delineated by the system manufacturer.
-  There is a risk to scanning feverish or decompensated cardiac patients.
-  Facial makeup should be removed before scanning because it may contain metal flakes which can cause skin and eye irritation. Permanent eyeliner tattoos may cause eye irritation due to ferromagnetic particles.
-  Patients who work in environments in which there is a risk of having embedded metallic fragments in or near the eye should be carefully screened before undergoing a MR exam.
-  Visually inspect the cable insulator jackets, strain reliefs, and connector boxes before each use. If the insulation is broken, or if the cable is frayed, immediately discontinue use of the device.

2-7 Emergency Procedures

In the unlikely event that a coil creates smoke, sparks or makes an unusually loud noise, or if the patient requires emergency assistance:

- Stop the scan if one is in progress.
- Remove the patient from the scan room if medical treatment is needed.

2-8 Technical Considerations

-  The coil and accessories require special conditions regarding electromagnetic compatibility. The coil must be installed and used in a shielded scan room provided with the MR system. The user must ensure that the scan room door is closed during system use. Failure to do so may cause reciprocal interference with portable and mobile RF communications equipment, affecting the performance of the MR coil and/or such equipment.
-  The coil should only be used with the accessories specified in the Operator's Manual.
-  The use of accessories other than those specified in the Operator's Manual may result in decreased ESD immunity of the coil or the system, causing damage to the coil and/or system.
-  The equipment should not be used with other coils or equipment present in the MR scanner except as specified in the Operator's Manual.
-  Tampering with the cable pins and connector may damage the connector and affect coil or system performance. Please verify that connector and pins are not damaged before use.
-  To avoid injuring the patient or damaging the coil, do not use non PET/MR mirrors #7 & #8 in PET/MR system.
Only use PET/MR mirrors #14 & #15 in PET/MR system.
-  Coil tilt is intended for use only while positioning the patient. Return the coil to the horizontal position before moving table. Advancing the table into the bore while the coil is tilted and when using mirrors/fMRI goggles can result in eye injury and/or damage the coil.

CHAPTER 3 – QUALITY ASSURANCE

Perform the system level Signal to Noise Check. Refer to Service Methods CD; System Level Procedures; Functional Checks; Signal to Noise Check.

3-1 Tools Required

TOOLS REQUIRED – TABLE 3-2-1

Description	GE Part #	Invivo #	Qty
18cm Diameter Spherical TLT Phantom w Silicon Oil	2359877	N/A	1
Phantom Positioner	2317112-3	102558	1

3-2 Setup

1. Attach the base plate with 3T 8 Ch Brain Coil to the cradle in the same position on the cradle as the standard head coil. Note the serial number of the coil.
2. Slide the coil back to permit access to the head holder as shown in Figure 3-2-2-1.
3. Place phantom ring on head holder as shown in Figure 3-2-2-2. The edge of the ring that goes toward the magnet has edges that overhang the head holder to secure the ring in place.
4. Place head TLT sphere on the phantom ring as shown in Figure 3-2-2-3.
5. Slide the coil all the way forward (away from the magnet) on the base plate and set the locks as shown in Figure 3-2-2-4.
6. Establish an axial landmark on the line on top of the coil. Advance to scan.

Figure 3-2-2-1



Slide Coil Back to Access Head Holder

Figure 3-2-2-2



Place Phantom Ring on Head Holder

Figure 3-2-2-3



Place Head TLT Sphere on Phantom Ring

Figure 3-2-2-4



Sphere Aligned with Coil

3-3 Multi-Coil Quality Assurance (MCQA) Tool

WARNING!

All RF coil related tests must be run on a system that is well-calibrated and passing all system tests, especially white pixel, correlated noise, and MCR (Multi-Coil-Receive) tool.

Hearing protection is required for all people in the scan room during a scan to prevent hearing impairment. Acoustic levels may exceed 99 dB(A). Hearing protection must have a Noise Reduction Rating (NRR) of 29 dB or better (e.g., 30 dB, 32 dB, etc.).

A typical testing procedure is comprised of the following steps:

- From the Service Browser, Image Quality Tab, select “Multi-Coil QA Tool” from the list of available tests.
- On the test side select “Click here to start the tool.”
- You will see a window with a list of available coils (Figure 1).

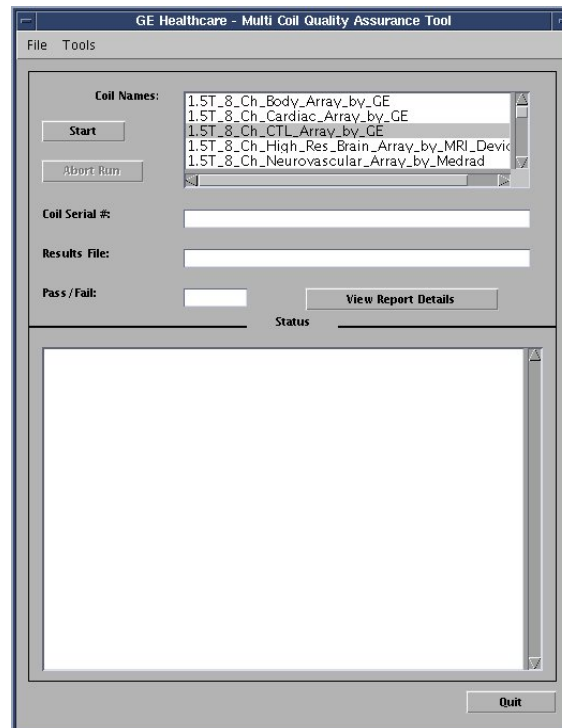


Figure 1: MCQA starting window.

- Choose the name of the coil under test. The list shown is just an example. Your site will have a different list based on the availability of the coils and field strength.
- Enter Serial number of coil.
- Click on START. This will initiate the automated test. Depending on the number of test locations (i.e complexity of the coil) the test may take from 3 to 20 mins.
- After the test is completed, you will see another window with the test results (Figure 2).

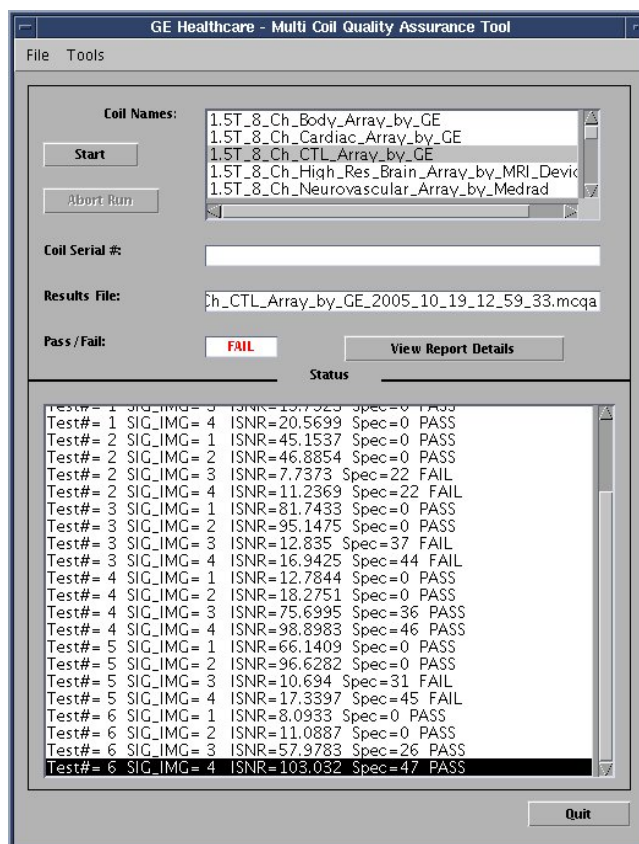


Figure 2: Output reported with PASS or FAIL for entire coil and for each test and channel

The results window shows Integrated Signal-to-Noise Ratio (ISNR) as a function of receiver number. The PASS/FAIL status should show PASS if all coil elements are functioning well.

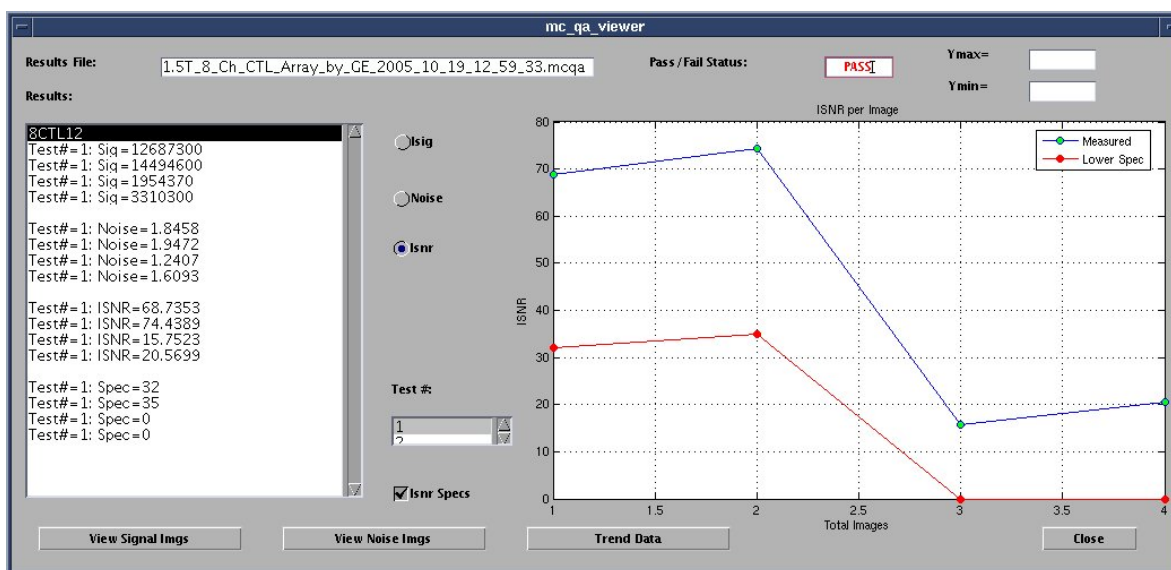


Figure 3: Window shown when "View Results Details" button is pressed. ISNR, Noise, or Integrated Signal is plotted as a function of receiver number

Upon start-up of the MCQA tool or during execution you may see the following messages:

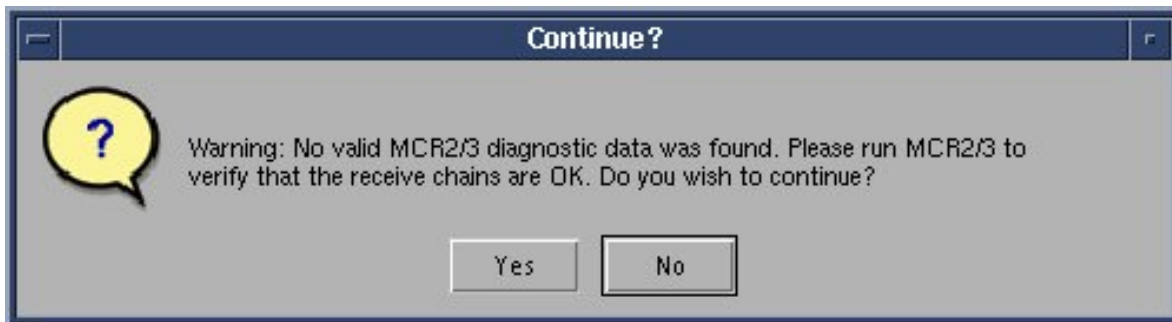


Figure 4: Missing MCR2/3 diagnostic data warning message

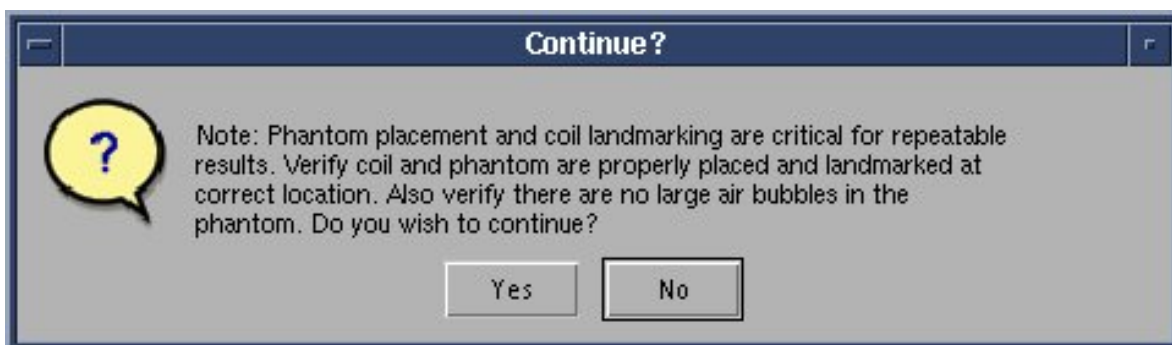


Figure 5: Phantom placement and landmarking message

CHAPTER 4 - USING THE 3T 8CH BRAIN COIL

Following is an example for using the coil in the GE 3T MR system. Refer to applicable System Operator Manual and System Service Manual Direction for use of the coil in the system.

4-1 Positioning the 3T 8CH BRAIN COIL

The 3T 8CH BRAIN COIL is designed for imaging of the entire head.

Use adapter to position the coil if the patient cradle does not match with bottom curvature of 3T 8CH BRAIN COIL.

The 3T 8CH BRAIN COIL must be placed on the patient cradle (or adapter) with the cable into the magnet. **The 3T 8CH BRAIN COIL is designed for head first exams only!**

3T 8CH BRAIN COIL base is designed to rest directly on the patient cradle (or adaptor). Standard GE patient cradle pads fit flush with the end of the 3T 8CH BRAIN COIL base.



Above: Insert the tabs on the coil base into the slots on the patient table (or adapter).

Right: Use the dimple (if available) to locate the coil base on the patient table or adapter.



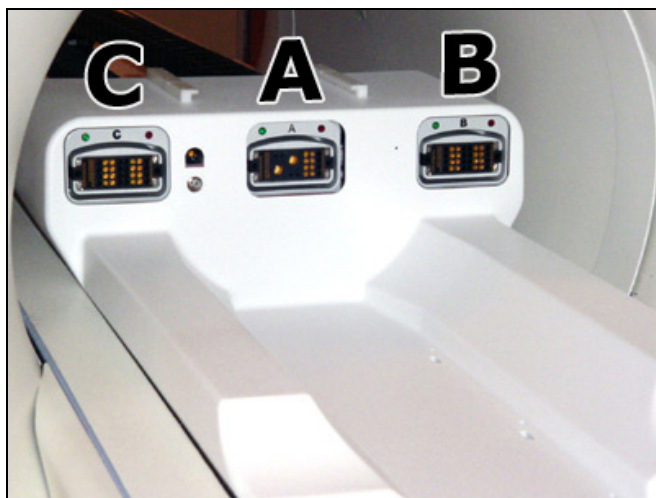
The patient comfort pad should be centered on the head holder. The pad serves to provide additional patient comfort during the scan, helps center the patient in the coil, and is recommended for optimal coil performance.



4-2 Connecting the Cable

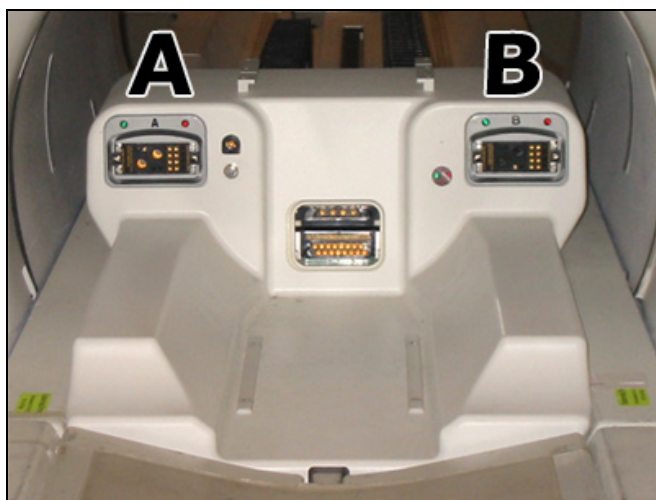
HDx

THE COIL WILL WORK IN
CONNECTOR A ONLY.



HDx Upgrade

THE COIL WILL WORK IN
CONNECTOR A ONLY.



Discovery MR750w with GEM Table
THE COIL WILL WORK IN
CONNECTOR A ONLY.



Discovery MR750
Discovery MR750w
SIGNA PET/MR

THE COIL WILL WORK IN
CONNECTOR A ONLY.



4-3 Positioning the Patient

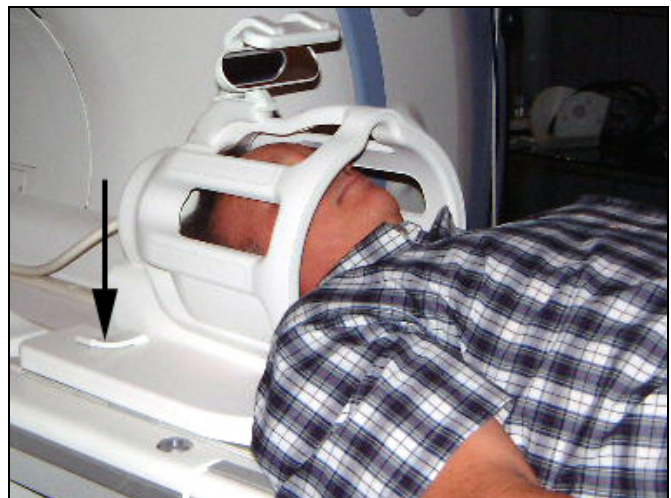
Place the patient's head on the head holder.



Slide the coil over the patient's head until it touches the shoulders. If necessary, insert wedge pads to help immobilize the patient's head.



Press down the locking levers to prevent the coil from moving during scanning.



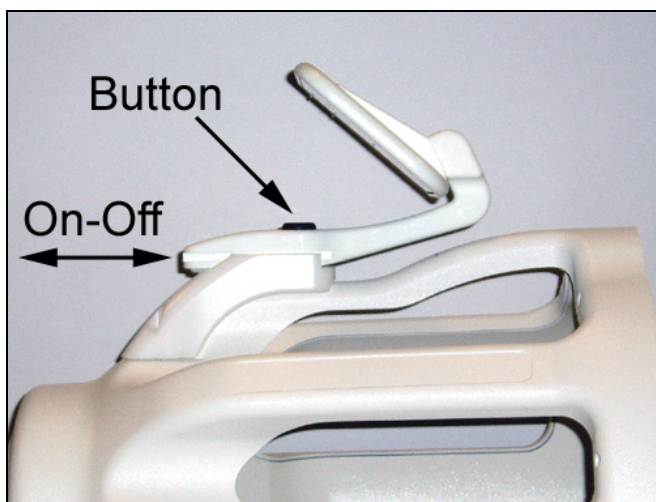
4-4 Using the Mirrors

There are two mirrors included with the 3T 8CH BRAIN COIL:

- Superior Mirror [right]
- Inferior Mirror [below]

To install either mirror, align the grooves in the base of the mirror with the grooves in the coil mirror mount, and slide the mirror into position.

Press and hold the release button on the mirror before attempting to remove the mirror.



The Superior Mirror [above] provides the patient with a view that is superior to the patient. The Inferior Mirror [right] provides the patient with a view that is inferior to the patient. In either case, the mirror is positioned by sliding until the patient has a clear view.



4-5 Establishing the Patient Landmark

Use the windows on the coil housing to ensure the patient is centered properly. Landmark on the anatomy of interest and advance the patient into the magnet for scanning.

4-6 Field of View and Coverage

For head studies an FOV of 24 cm. is suggested, depending upon head size. The High Res Brain Array is 24 cm. in diameter. A smaller FOV may be used if desired.

4-7 Patient Hearing Protection

Provide ear plugs for the patient after all instructions have been given.



Hearing protection is required for all people in the scan room during a scan to prevent hearing impairment. Acoustic levels may exceed 99 dB(A).

Hearing protection must have a Noise Reduction Rating (NRR) of 29 dB or better (e.g., 30 dB, 32 dB, etc.).

CHAPTER 5 - SCANNING

5-1 Selecting the Coil

The coil mode that is available for use is 8HRBRAIN for MR System. The coil description displayed to the user is **HD 8Ch High Res Brain Array by Invivo** for MR System.

The coil mode that is available for use is 8HRBRAIN for PET/MR System. The coil description displayed to the user is **PETMR 8Ch High Res Brain Array by Invivo** for PET/MR System.

This coil mode uses all 8 elements of the coil.

5-2 Auto Pre-scan

Generally, the system auto pre-scan will function normally when used with the 3T 8Ch Brain Coil. If auto pre-scan returns the message "auto pre-scan failed to complete normally," then manual pre-scan may be required to acquire the image. Consult the system operator's manual for manual pre-scan procedures.

5-3 Localizing with the Body Coil

The system body coil may be used at any time while the 3T 8Ch Brain Coil is in the scanner and connected to the system port. This allows a large FOV (48 cm.) body coil localizer to be performed, if desired.

5-4 Using Autoshim

AUTOSHIM is a feature of the AUTOPRESCAN software that improves image quality. It does this by improving the magnetic field homogeneity within the FOV selected. The improvement in image quality is often dramatic when the selected FOV is far off center, and when acquiring FSE or T2 weighted scans, or FAT SAT images.

5-5 Magnetization Transfer Techniques

Magnetization transfer techniques are available with GE Angio software.

5-6 Protocols

Refer to the GE protocols on your system.

CHAPTER 6 – INSTALLATION AND MAINTENANCE

6-1 Installation and Configuration

Coil installation and configuration requires the services of your GE Service Representative. Please refer to the service methods DVD for coil installation and configuration information.

6-2 Cleaning

The 3T 8 Ch Brain Coil, base and pads must be cleaned and stored using the following procedure:

- Wipe with a cloth that has been dampened in a solution of 10% bleach and 90% tap water, or 30% isopropyl alcohol and 70% tap water.
- **Do not pour any cleaning solution directly on the coil!**
- Let the coil housing and pads dry before use.
- **Under no circumstances should the coil be placed into any type of sterilizer.**

6-3 Storage

This equipment must be stored under the following conditions:

- Ambient temperature range of +10°C to +40°C
- Relative humidity range of 10% TO 90%, non-condensing
- Atmospheric pressure range of 500 hPa TO 1060 hPa

To store the coil, a storage space of 38.1 cm/15.0 in (width) x 46.0 cm/18.1 in (depth) x 35.6 cm/14.0 in (height) is required.

6-4 Replaceable Accessories

The following table lists replaceable accessories that may be purchased from Invivo Corporation:

Description	Invivo #
Thick Head Pad	870305
Thin Head Pad	870306
Spacer Pad	870307
Wedge Pad	870308

6-5 Contact Information

For additional information, contact:

GE Healthcare Americas
USA 800-558-5102
Canada 800-668-0732

GE Healthcare Europe
(33) 1-41-19-76-76

GE Healthcare Web Site
www.gehealthcare.com

GE Healthcare Asia
China 86-21-62192228
Taiwan 886-2-2505-7900
Singapore 65-291-8528
Australia 61-2-9975-5501
Japan 81-120-48-2630
Korea 82-31-740-6119
India 91-80-845-2923

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